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A novel concrete-filled auxetic tube composite structure: Design and compressive characteristic study

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ABSTRACT

Concrete-filled stainless steel tube (CFSST) members take advantage of the high strength and the outstanding corrosion resistance to act as an important role in civil engineering structures. However, the steel tube could not provide the perfect confinement effect for the core concrete during the initial elastic compression stage because Poisson's ratio of the concrete is smaller than that of the stainless steel tube (SST). In this paper, a novel concrete-filled auxetic stainless steel tube (CFASST) composite structure was designed and manufactured to actively restrain the concrete, making the best use of the desirable deformation characteristics of auxetic tubular structures. The axial compressive performance of these CFASST members and their control factors were investigated experimentally and numerically. Test results were discussed in detail which included failure modes, load versus displacement curves and strain analysis. Finally, parametric analyses were conducted to further study the effects of different parameters (Poisson's ratio, thickness of the stainless tube) on the CFSST composite structure under axial compression. It was found that CFASST composite structures possess an unusual deformation mode and an improved confinement effect.

1. Introduction

Many approaches have been developed to improve the toughness of concrete, such as fiber-reinforced concrete [1–3] and concrete-filled steel tubes (CFSTs). CFST members play an important role in civil engineering structures because of high bearing capacity, favorable ductility [4] and excellent energy absorption capacities [5]. A great number of researchers have been attracted to investigate the mechanical properties of CFST members in the past decades [6–8]. However, traditional CFST members suffer from corrosion problems, especially for offshore or underground construction. Consequently, anti-corrosion measures are taken to maintain structural durability, resulting in increased maintenance costs. In contrast with carbon steels, stainless steels own prominent benefits of high anti-corrosion performance, better durability, and fire resistance [9]. To this end, it can be considered as an effective method to solve the corrosion problem that concrete-filled stainless steel tube (CFSST) members combine the high corrosion resistance of stainless steel with the advantages of CFST members. Systematic researches on CFSST members have been extensively conducted recently [10–13]. Nevertheless, as reported by Lai et al. [14], the steel tube could not provide the perfect confinement effect for the core concrete during the initial elastic compression stage because Poisson's ratio of the stainless steel tube (SST) is greater than that of the concrete. Imperfect interface bonding results in a decrease in the initial strength and stiffness of the CFST members. It was found that interface bonding effects of CFSST members are related to concrete strength [15], steel tube thickness [16], especially Poisson's ratio of the materials. In addition, outward local buckling in the tube can be found for the CFST columns whose failure modes lead to a sudden reduction in the load-bearing capacity of the CFST members, thereby threatening the safety of high-rise buildings.

Auxetic materials represent materials and structures with negative Poisson's ratio (NPR), exhibiting unusual deformation characteristics that contract (expansion) transversely when subjected to uniaxial compression (tensile) load [17,18]. Depending on this unusual behaviour, auxetic metamaterials are endowed with various superior mechanical properties [65–70], such as in-plane indentation resistance [19,20], shear resistance [21,22], synclastic behaviour [23], energy

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absorption capability [24,25,59], bending resistance [26]. Therefore, preliminary applications of auxetic structures have extended to many fields, including medical equipment, intelligent materials, acoustic engineering [71,72], civil engineering [27–29,63]. So far, various auxetic structures have been systematically investigated, e.g., re-entrant structures [30,62], rigid rotation structures [31,32], chiral structures [33,34,61,64], buckling-induced structures [35–38], *etc*. In particular, as a special part of auxetic structures, auxetic tubular structures require further research with the intention of capitalizing on the excellent auxetic behaviour for multi-functional applications [39–41,60].

Auxetic tubular structures exhibit multifunctional and unusual performance, such as impact resistance performance [42,43], bending performance [44] and synclastic behaviour [45]. Based on this, widely studies have been implemented on auxetic tubular structures towards exploring their applications as adjustable medical and energy-absorbing devices. Scarpa et al. [46] laid out a pioneering study on analytical and numerical modeling of auxetic tubular structures composed of re-entrant hexagonal honeycomb elements. Grima et al. [45] proposed a semi re-entrant honeycomb cylinder and indicated that it has a natural tendency to form tubular structures. Bhullar et al. [47] adopted rotating rigid square units to design and manufacture auxetic metal and polymer stents, using advanced manufacturing technology. Ren et al. [36] designed buckling-induced metallic tubular structures whose mechanical properties can be tuned by adjusting the value of the pattern scale factor. Limited investigations listed above showed that auxetic tubular structures are able to produce obvious radial shrinkage under axial compressive load, which is beneficial to improve mechanical properties. However, when the slenderness ratio and width-to-thickness ratio of auxetic tubular structures exceed the critical value, it will trigger global or local buckling modes [48]. It has been found that previous studies mainly concentrated on mechanical performance analysis of a single auxetic tube and few publications have been reported to study the auxetic tubular composites acting as a solution to prevent the auxetic tube from buckling.

In order to solve the problem that the SST cannot effectively restrain concrete in the initial elastic phase and the SST develops the local buckling bulge, resulting in imperfect interface bonding. A novel concrete-filled auxetic stainless steel tube composite structure was proposed to restrain concrete actively which makes full use of the auxetic behaviour characteristic that auxetic tubular structure contracts radially inward under axial compression. At the same time, the local buckling phenomenon is improved due to the support of the concrete core. To this end, three types of specimens made of SSTs which are manufactured employing laser cutting technology were tested under compressive load to estimate their compressive bearing capacities and failure modes. Finite element analysis methods were used to validate against the experimental results, and the effects of different parameters on the compressive performance were discussed in the parametric study. Influences of Poisson’s ratio value and the thickness of the SST on the compressive behaviour were studied based on the numerical analysis results. Finally, the confinement effect of different SSTs on the concrete was discussed through numerical analysis.

2. Design and fabrication of auxetic and non-auxetic stainless steel tubes

2.1. Design of auxetic and non-auxetic stainless steel tubes

Auxetic tubular structures are designed through the coordinate transformation approach that convolves 2D auxetic structures made of periodically arranged elliptic unit cells into 3D auxetic tubes, and extending along with radius directions, as shown in Fig. 1. Based on our previous works of auxetic tubes [36], three groups of SSTs were designed and the geometrical configurations of them are shown in Fig. 2. The three groups of SSTs included an auxetic SST (ASST) with alternatingly ellipse holes, a non-auxetic SST (NSST) with circular holes and a traditional non-auxetic SST (TSST) without holes. As shown in Fig. 3(a), the main parameters of the unit cell contain the vertical and horizontal length L, the thickness t of the ribs, the long-axis a and the short-axis b of the ellipse holes. The specific parameters for the two kinds of unit cells are listed in Table 1. The geometric parameters of these SSTs are shown in Fig. 3(b), where H is the height of the SST, D is the outer diameter of the SST, T is the wall thickness of the SST. The specific parameters for the three groups of SSTs are listed in Table 2. For comparison, the number and arrangement mode of holes and superficial area of the holes between ASSTs and NSSTs were kept the same. In order to ensure that the designed ASST has obvious auxetic behaviour, its porosity is set to 41.1% [49]. It should be noted that the NSST does not exhibit auxetic behaviour in that when the base material is metal, the buckling-induced mechanism is not applicable [35].

2.2. Fabrication of auxetic and non-auxetic CFSST composite structures

The laser cutting technology was adopted to fabricate the designed three types of SSTs and the base material was 304 stainless steel. Before casting the concrete, a 304 stainless steel plate was welded onto the bottom end of the steel tube. In order to prevent concrete from flowing out of the tube walls through the holes, polyurethane foams mechanically cut were employed to plug these holes and stretch film with viscosity was stuck to the tube wall. Then, the concrete was poured into the tubes and the specimens were completely compacted through a vibration table. After the initial setting of concrete, the stretch film was torn off and the polyurethane foam was taken out. In order to reduce the adverse effects of the concrete shrinkage, the high-strength mortar was then employed to fill in the gap between the tube and the concrete after curing the concrete for 28 days. Subsequently, the concrete sander was adopted to polish the concrete surface. CFSST composite structures’ production process was shown in Fig. 4. Besides, these CFSST composite structures are divided into three categories according to different types of tubes: concrete-filled auxetic stainless steel tube (CFASST) composite structures, concrete-filled non-auxetic stainless steel tube with circular holes (CFNSST) composite structures and concrete-filled traditional

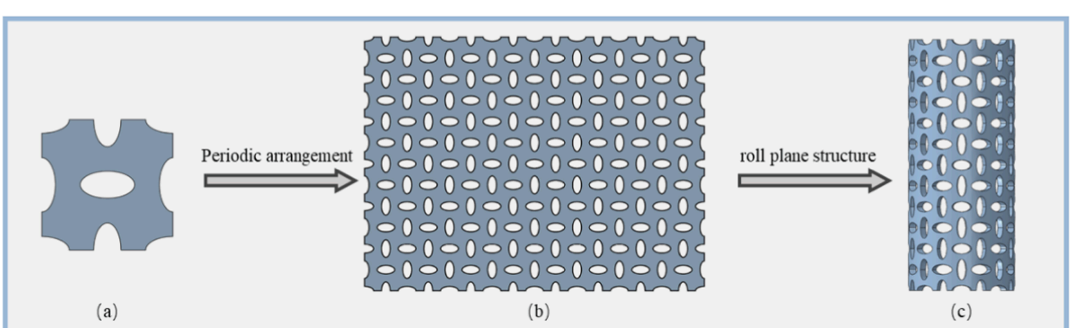

Fig. 1. Coordinate transformation approach for auxetic tubular structures design: (a) typical unit cell; (b) 2D auxetic plane structure; (c) 3D auxetic tubular structure.

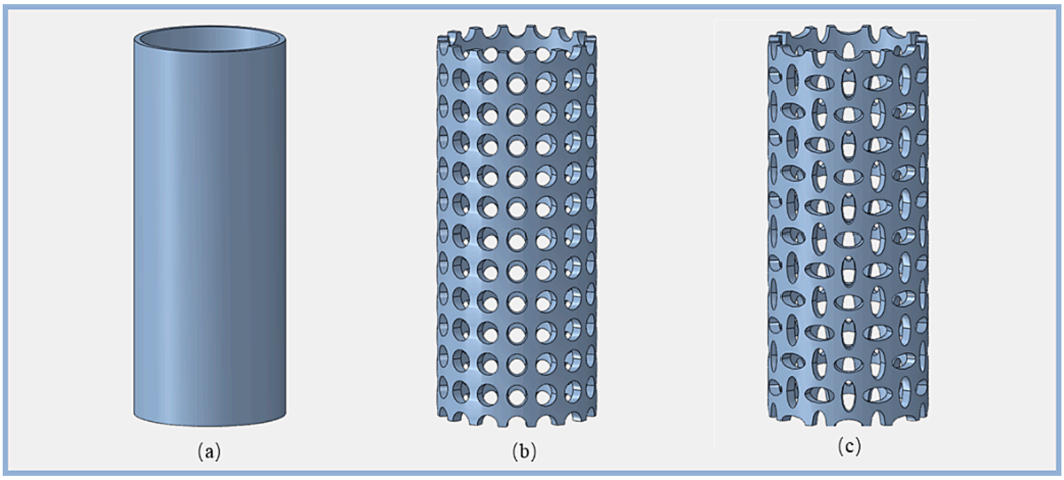

Fig. 2. Geometrical configurations of three types of SSTs: (a) TSST; (b) NSST; (c) ASST.

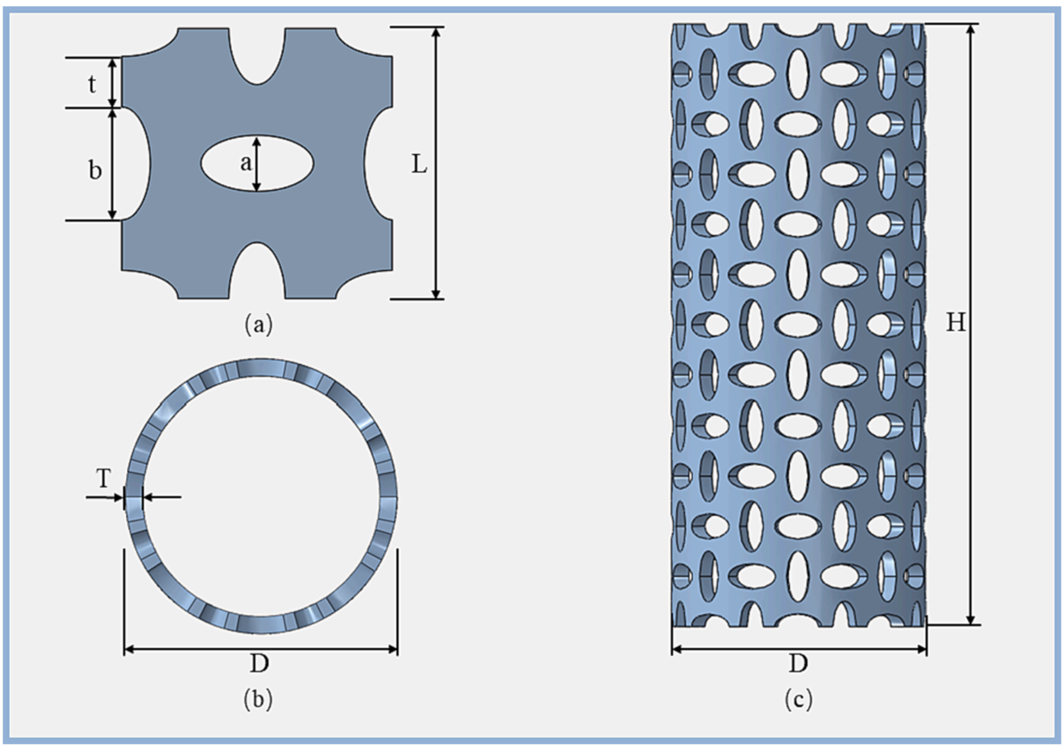

Fig. 3. Geometric parameters of ASST: (a) cell schematic diagram; (b) top view; (c) front view.

Table 1
List of unit cell parameters.

Unit cell label	*L* (mm)	*t* (mm)	*a* (mm)	*b* (mm)
NSST	62.4	8.8	22.4	22.4
ASST	62.4	7.5	31.6	15.8

Table 2
Geometric and property parameters of CFSST specimens.

Specimen label	*D* (mm)	*T*(mm)	*H* (mm)	$f_{0.2}$(MPa)	f_{cu}(MPa)	α
CFTSST	152.1	6.58	373.5	261	46.5	0.15
CFNSST	158.3	10.46	373.5	261	46.5	0.15
CFASST	158.3	10.37	373.5	261	46.5	0.15

stainless steel tube without holes (CFTSST) composite structures. The material strength properties and geometric parameters of all specimens are listed in Table 2, where $f_{0.2}$ is the yield strength, f_{cu} is the compressive concrete cube strength and α is the steel content of the test specimens.

3. Experimental scheme

3.1. Material test

The tensile coupon tests were executed to determine the properties of stainless steel material [50]. Typical stress-strain curves of the 304 stainless steel are shown in Fig. 5. Actual measured results indicate that the material curves have no clear yield platform. The material properties of the stainless steel are given in Table 3, where *t* is the thickness, f_u and

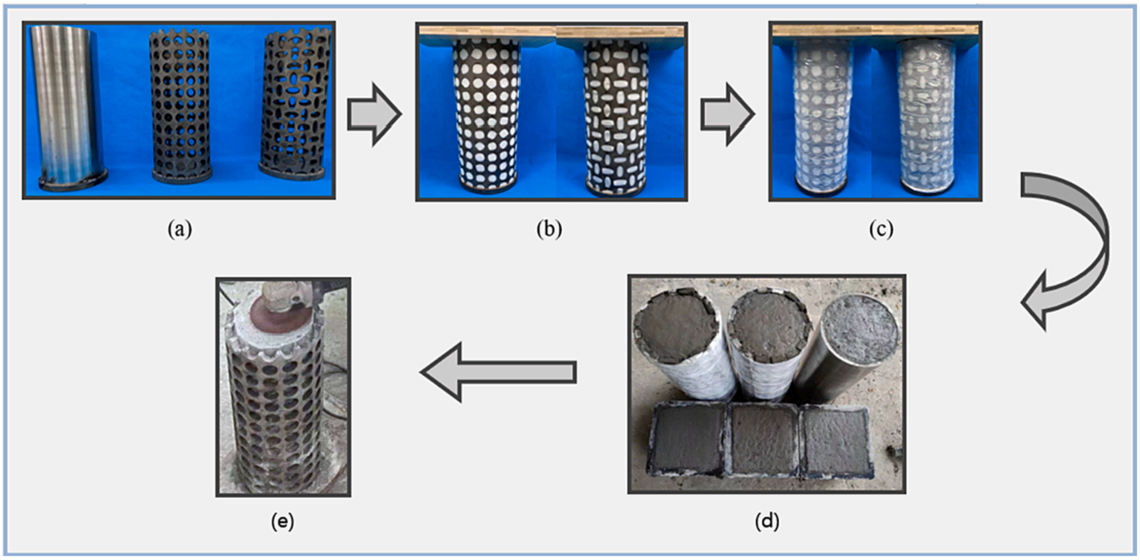

Fig. 4. Fabrication process of CFSST composite structures: (a) laser cutting technology; (b) employ foam to plug holes; (c) stick film to tube wall; (d) curing the concrete; (e) polish the concrete surface.

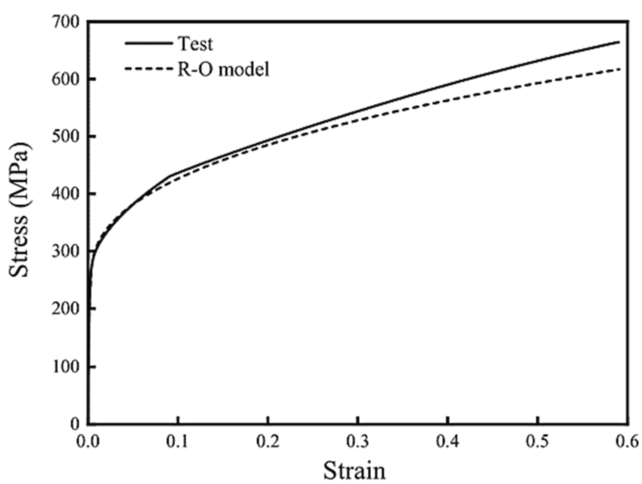

Fig. 5. Stress-strain curves of the stainless steel.

Table 3
Measured material properties of the stainless steel.

Material	t (mm)	$f_{0.2}$(MPa)	f_u(MPa)	E_0(MPa)	ε_u(%)
304 austenitic stainless steel	6.0	261	654	1.88×10^5	62.3

ε_u are the ultimate stress and the ultimate strain, E_0 is Young's modulus.

According to Chinese standard GB 50081–2011 [51], cube strength tests were carried out to determine concrete material properties. Three 150 mm × 150 mm × 150 mm concrete cube specimens were tested to obtain cube strength. The mean compressive strength of the concrete cube was 46.5 MPa.

3.2. Test setup

A description of the test setup is shown in Fig. 6 (a). An electric 5000 kN hydraulic testing machine was employed to conduct the quasi-static compressive tests on the specimens. The indenter moved axially at a speed of 1 mm/min which is corresponding to the quasi-static compression. Compression experiments were ended when the displacement reached about 50 mm. The axial displacement was measured using 2 linear variable displacement transducers (LVDTs) which were placed on either side of the members. Four pairs of strain gauges (SG) were arranged at 90° intervals circumferentially at the mid-height, one-third and two-thirds height cross-section (Fig. 6 (c)).

4. Experimental results

4.1. Failure modes and process

Failure modes of the CFSST specimens are shown in Fig. 7. At the initial loading stage, three CFSST composite structures had no obvious deformation, however, all the specimens occurred slightly bending because of the inclined load caused by the machine. As the axial load started to increase, obvious outward local buckling could be observed for CFTSST composite structures and developed quickly, as shown in Fig. 7 (a), where local buckling occurred near the top and bottom ends of the specimens. However, the deformation of CFNSST and CFASST composite structures was completely different from the former in that their continued expansion replaced the local buckling bulges due to the deformable characterization of the holes. It was obvious that the expansion in the middle was larger than that in the top and bottom sides. Moreover, the expansion and later deformation of CFASST composite structures were less than CFNSST composite structures.

4.2. Axial load versus displacement curves

Fig. 8 shows the load versus axial deformation relationships for three CFSST composite structures and it can be concluded that stainless tubes can effectively increase concrete members' strength and slow down the descending segment of the post peak curves. Generally, load-displacement curves can be separated into four stages of CFSST composite structures. The first stage is the initial stage of loading, three CFSST composite structures were in the elastic phase where CFTSST composite structures' compressive initial stiffness was the largest among three specimens. Initial stiffness is the slope of the load-displacement curve in the elastic stage. It is due to the fact that the stiffness of the TSSTs is much greater than that of the other two tubes. By comparing the other composite structures, the stiffness of the CFASST composite structure is almost the same as the CFNSST composite structure in the early stage of the curve. This phenomenon is caused by the active

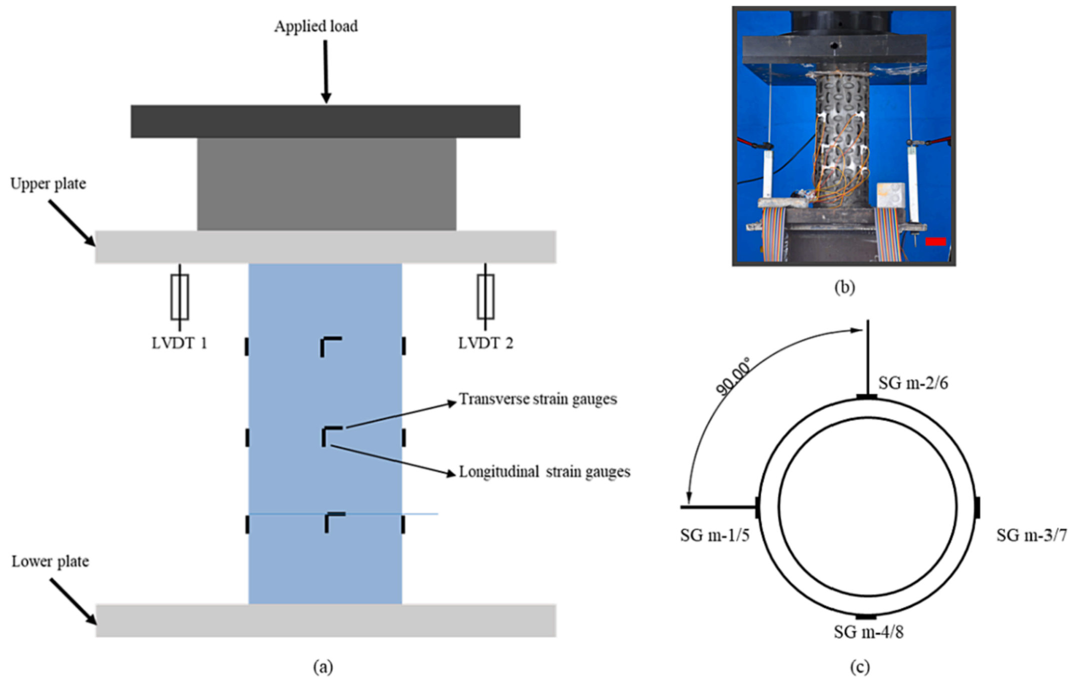

Fig. 6. The test setup of CFSST composite structure: (a) schematic view of test setup; (b) photo of test setup; (c) locations of strain gauges at middle section. (Scale bar: 50 mm.).

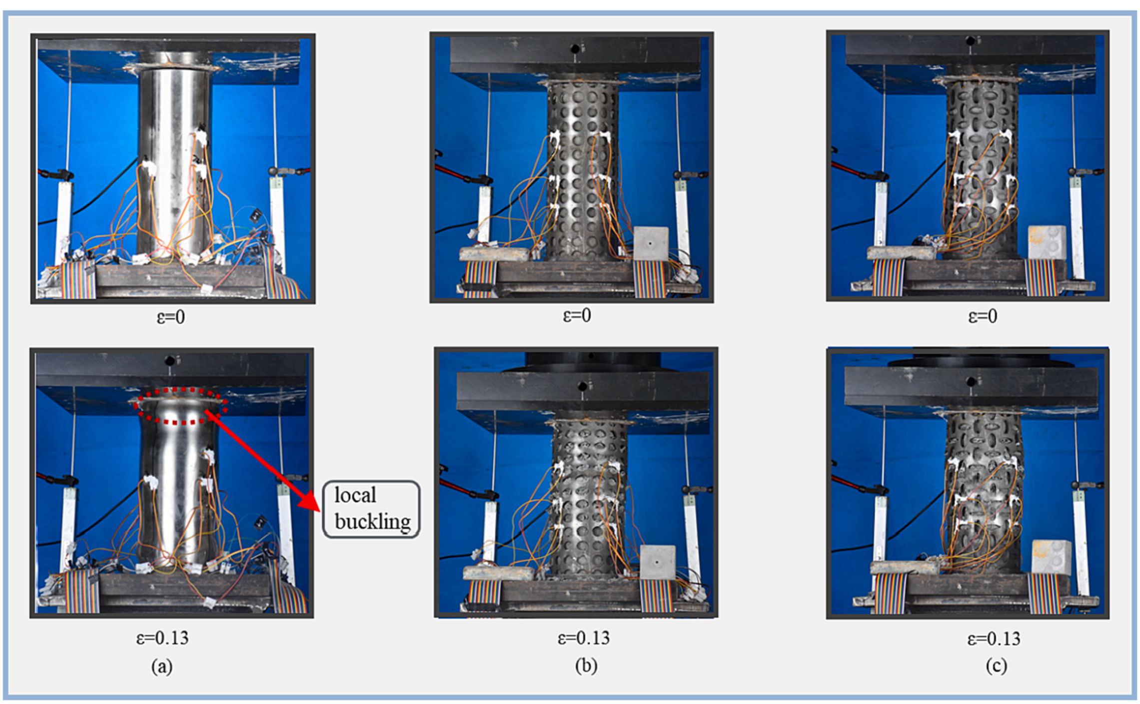

Fig. 7. Failure modes of the CFSST specimens: (a) CFTSST composite structures; (b) CFNSST composite structures; (c) CFASST composite structures.

restraint effect of the ASST on the concrete. The second stage is the elastic-plastic phase where the load reached its first peak. CFASST composite structures' peak load was higher than CFNSST composite structures and lower than CFTSST composite structures. This is because auxetic tubes with ellipse holes sacrifice structural stiffness and strength for achieving the auxetic behaviour. However, the CFASST composite structures' ductility and toughness are the best. The third stage is the decreasing phase of the loading and load-displacement curves of them were significantly different. On one side, the load-displacement curves of CFNSST composite structures decreased slower than that of CFTSST composite structures; on the other side, load-displacement curves of CFASST composite structures displayed an almost horizontal trend with no descending stage. The difference was caused by that the confining effect of the ASSTs on the concrete was higher than that of NSSTs. The last stage is the continuous increasing phase of the loading where CFTSST composite structures' load rose the fastest. It is due to the fact that most of the bearing capacity is provided by the stainless steel tube because the core concrete is damaged under the large compressive strain

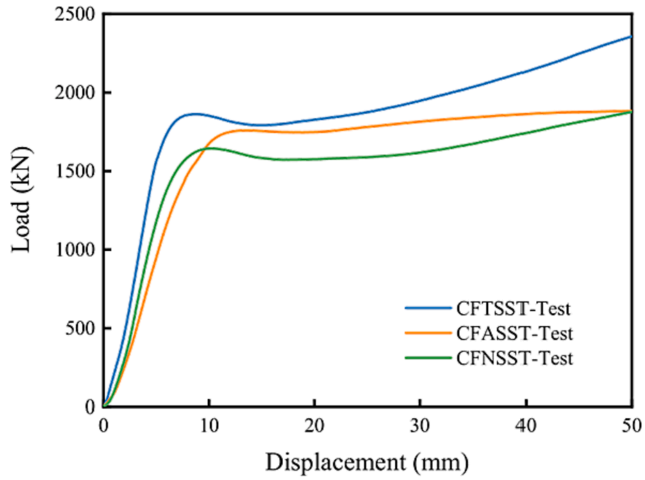

Fig. 8. Load-displacement curves of CFSST composite structures.

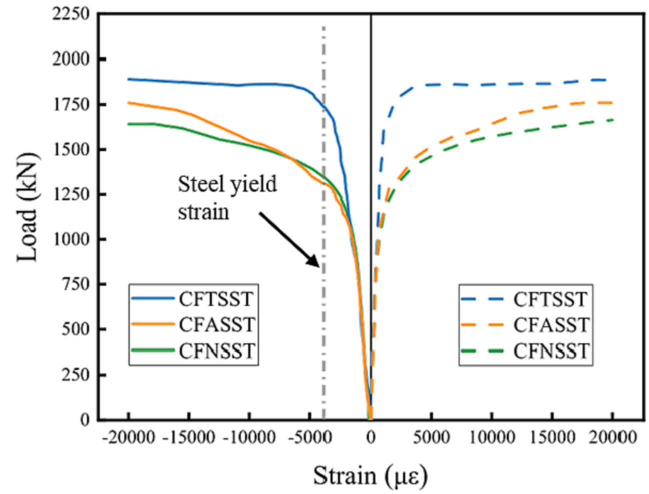

Fig. 9. Load-strain curves of CFSST composite structures.

in this stage. While the ASST's and NSST's bearing capacities are lower than that of the TSST. Furthermore, the stainless steel has no apparent yield point and the compressive load of the stainless steel continues to increase with the increasing displacement. The axial compressive performance of three CFSST composite structures are summarized in Table 4, where N_u is the experimental first peak load, Δ_u is the experimental first peak displacement, Δ_y is the yield displacement, K_i is the initial stiffness, μ is the ductility coefficient and $\mu = \Delta_u/\Delta_y$.

4.3. Strain analysis

Fig. 9 shows the load versus strains curves of three CFSST members. In general, the transverse strains developed slower than the longitudinal strains before reaching the peak load for all specimens. The load versus strain curves were nearly linear during the initial stage of loading and displayed strain-hardening performance along with the increasing load. Compared with the CFTSST composite structures, CFASST and CFNSST composite structures' peak loads occurred late and the strain development contribute to the continuous increase of the load. This phenomenon was attributed to the fact that CFTSST composite structures were prone to local buckling.

5. Numerical results

Commercial finite element analysis package Abaqus/Explicit solver was adopted to further explore the effects of various parameters on CFSST composite structures compressive behaviour and stainless steel tube confinement effect.

5.1. Material modeling of stainless steel tube

The main property material parameters of the SSTs were determined based on material test results, as reported in Table 3. The R-O material model was adopted to represent the stress-strain (σ-ε) curve in the finite element model [52,53].

Table 4
Experimental results for all specimens.

Specimen label	N_u (kN)	Δ_u (mm)	Δ_y (mm)	K_i(kN/mm)	μ
CFTSST	1862	8.83	6.54	311.04	1.35
CFNSST	1643	10.15	7.79	235.57	1.30
CFASST	1759	13.74	10.01	194.14	1.37

$$\varepsilon = \begin{cases} \dfrac{\sigma}{E_0} + 0.002\left(\dfrac{\sigma}{\sigma_{0.2}}\right)^n (\sigma \le \sigma_{0.2}) \\ \dfrac{\sigma - \sigma_{0.2}}{E_{0.2}} + \varepsilon_u\left(\dfrac{\sigma - \sigma_{0.2}}{\sigma_u - \sigma_{0.2}}\right) + \varepsilon_{0.2}(\sigma > \sigma_{0.2}) \end{cases} \tag{1}$$

$$n = \frac{\ln 20}{\ln(\sigma_{0.2}/\sigma_{0.01})} \tag{2}$$

$$E_{0.2} = \frac{E_0}{1 + 0.002nE_0/\sigma_{0.2}} \tag{3}$$

$$\frac{\sigma_{0.2}}{\sigma_u} = \frac{0.2 + 185\sigma_{0.2}/E_0}{1\text{-}0.0375(n-5)} \tag{4}$$

$$\varepsilon_{0.2} = \sigma_{0.2}/E_0 + 0.002 \tag{5}$$

$$\varepsilon_u = 1 - \sigma_{0.2}/\sigma_u \tag{6}$$

$$m = 1 + 3.5\sigma_{0.2}/\sigma_u \tag{7}$$

where $\sigma_{0.2}$ is the average 0.2% proof stress and E_0 is the initial elastic modulus; $\sigma_{0.01}$ and n are the measured 0.01% proof stress and the strain-hardening exponent determined; $E_{0.2}$ is the tangent elastic modulus when the stress reaches $\sigma_{0.2}$; σ_u is the ultimate stress formulated [28]; $\varepsilon_{0.2}$ and ε_u are the strain at 0.2% proof stress and the ultimate strain, respectively. The R-O model material curve was drawn in Fig. 5.

It should be noted that "nominal stress" and "nominal strain" were converted to "real stress" and "real strain" when the stress–strain relationship was incorporated into the FE models using Eqs. (2) and (3), respectively.

$$\sigma_{\text{true}} = \sigma_{\text{nom}}(1 + \varepsilon_{\text{nom}}) \tag{8}$$

$$\varepsilon_{\text{true}} = \ln(1 + \varepsilon_{\text{nom}}) \tag{9}$$

5.2. Material modeling of concrete

The concrete damaged plasticity model was determined based on the Refs. [54–55], which mainly includes three aspects: the compressive behaviour, the tensile behaviour, and the plasticity parameters. The elastic modulus was calculated according to Eq. (10) and the Poisson's ratio is 0.2 [56]. The key plasticity parameters were defined following the approach in Ref. [55]. The compressive behaviour of confinement concrete was represented by the axial stress-strain curve according to [54], as shown in Eq. (11). The crack energy model proposed in [57] was adopted to simulate the tensile behaviour of the concrete using Eq. (12). The tensile and compressive behaviours of the confined concrete were described by stress-strain curves, as shown in Fig. 10(a) and (b).

$$E_C = 4700\sqrt{f_c^{'}} \tag{10}$$

$$y = \begin{cases} 2x - x^2 (x \leq 1) \\ \dfrac{x}{\beta_0 (x-1)^2 + x} (x > 1) \end{cases} \tag{11}$$

where $x = \varepsilon/\varepsilon_0$, $y = \sigma/\sigma_0$, ε is the compressive strain and σ is the compressive stress; $\sigma_0 = f_c^{'}$ is the cylinder compressive strength of concrete; $\varepsilon_0 = \varepsilon_c + 800\xi^{0.2} \times 10^{-6}$; $\varepsilon_c = \left(1300 + 12.5 f_c^{'}\right) \times 10^{-6}$; $\beta_0 = \left(2.36 \times 10^{-5}\right)^{[0.25+(\xi-0.5)7]} \left(f_c^{'}\right)^{0.5} \times 0.5$; $\xi = A_S f_{0.2}/A_C f_{ck}$.

$$y = \begin{cases} 1.2x - 0.2x^6 (x \leq 1) \\ \dfrac{x}{0.31\sigma_p^2 (x-1)^{1.7} + x} (x > 1) \end{cases} \tag{12}$$

where $x = \varepsilon_t/\varepsilon_p$, $y = \sigma_t/\sigma_p$; ε_t is the tensile strain and σ_t is the tensile stress; σ_p is the peak tensile stress and ε_p is the peak tensile stain; $\varepsilon_p = 43.1\sigma_p(\mu\varepsilon)$; $\sigma_p = 0.26\left(1.25 f_c^{'}\right)^{2/3}$.

5.3. Validation of finite element model

Two analytical rigid plates were placed at both ends of the CFSST composite structure models to simulate the punch and plate in the experiment (Fig. 11). General contact was applied to the whole system during the loading process. The hard contact was adopted in the normal direction and the friction coefficient was set to 0.25 in the tangential direction. The bottom plate was fixed on the concrete and the bottom of the tube using a tie constrain. The top plate was set to a reference point and a vertical point load was applied to the top plate. The top rigid plate compression velocity was 1 mm/min, which is the same as the experimental condition set. Solid elements with reduced integration (C3D8R) were adopted to build the model and the mesh sweeping seed size of the concrete was 5 mm, the mesh sweeping seed size of the auxetic tube was 2.5 mm. Element convergence analysis was conducted during the FE analysis.

Comparisons of numerical and experimental results are shown in Fig. 12, including the failure modes and load-displacement curves. Good agreement was confirmed between test results and FE analysis results. Local buckling that occurred in the experiment process could be accurately simulated by the FE analysis for CFTSST composite structures. Similarly, continuous expansion behaviour and the densification process of holes can also be exactly captured in the FE models of the CFASST and CFNSST composite structures. The load-displacement curves of experiment and finite element have a 10% error, which illustrates the results of CFSST specimens between simulation and experiment match well. Generally, the load calculated finite element analysis is higher than that of the test for the reason that the concrete-filled into the steel tube could not be completely dense. In addition, laser cutting technology might

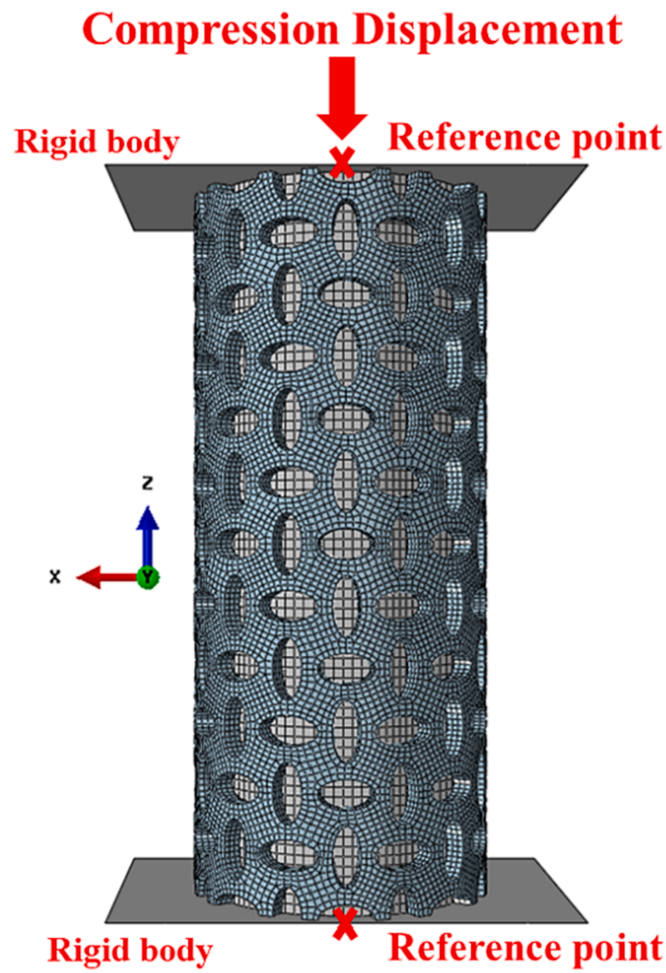

Fig. 11. Finite element model of CFASST composite structure.

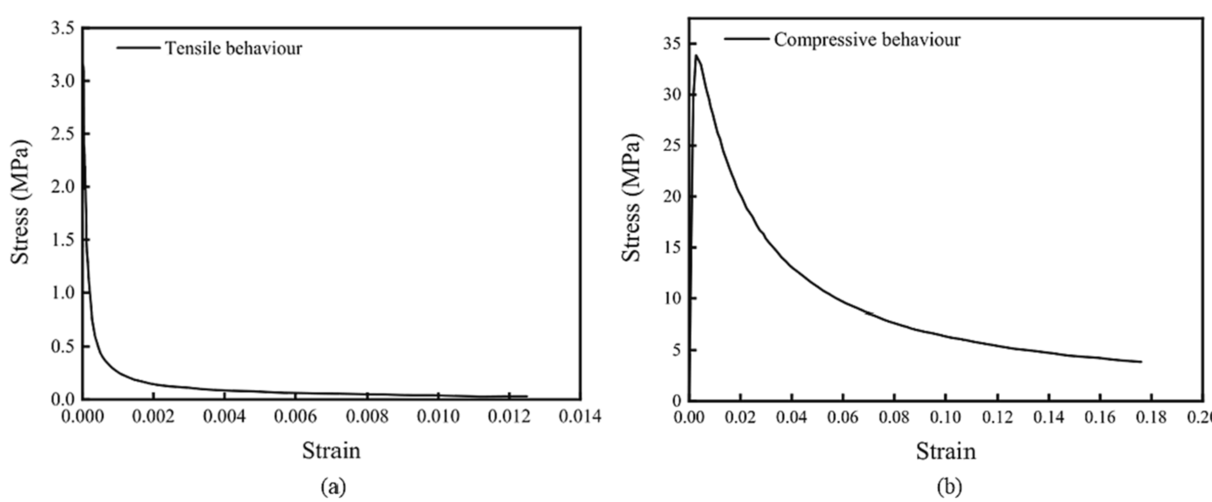

Fig. 10. Stress-strain curves of the confined concrete.

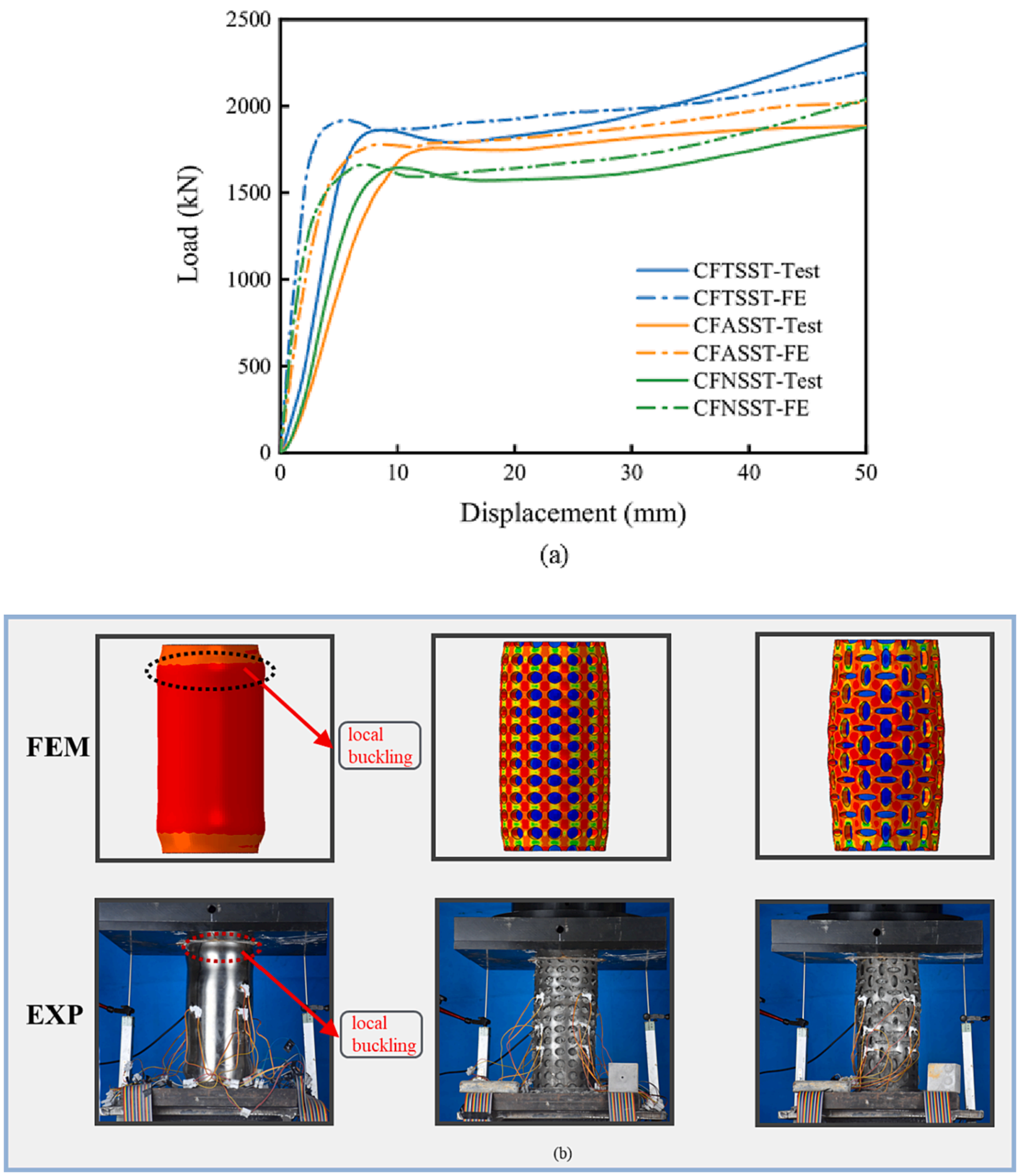

Fig. 12. Comparisons of numerical and experimental result: (a) load-displacement curves; (b) failure modes.

cause some damage to the stainless steel, but the FE model is perfect which will lead to the difference between experimental and simulated results. However, the peak load and trend of the load-displacement curves still remain the good agreement between test results and FE analysis results, so the numerical simulation results are relatively worthy of reference.

6. Parametric study and discussion

In order to further explore the effects of different parameters (Poisson's ratio of the stainless tube, thickness of the stainless tube) on the CFSST composite structures under axial compression, a series of parametric analyses were carried out using Abaqus based on the validation of finite element models. Besides, the SST confinement effect on the concrete was investigated through numerical analysis.

6.1. Influence of Poisson's ratio of the stainless steel tube

In order to investigate the effect of Poisson's ratio on the compressive strength of CFSST composite structures, three samples with different Poisson's ratios were designed and simulated. The Poisson's ratio value was set to −0.29, 0 and 0.29, respectively. Fig. 13 shows load-displacement relationships of CFTSST FE models with varying Poisson's ratios. It can be found that as Poisson's ratio decreases from 0.29 to −0.29, the compressive bearing capacities of CFSST composite structures increase significantly in terms of both the peak load and the initial stiffness. It is attributed to the confinement effect provided by ASST that will restrain concrete in the whole compressive process, resulting in favorable interface bonding between concrete and stainless steel tube. The results indicate that decreasing the Poisson's ratio of the stainless steel tube is beneficial for improving the compressive performance of CFSST composite structures. However, it is worth noticing that natural auxetic material has been found rarely. To the best knowledge of authors, auxetic tubular structures could only be achieved through structural design at the cost of sacrificing structural strength. Therefore, designing tubular structures both have a large auxetic behaviour and the improved mechanical property is crucial and indispensable.

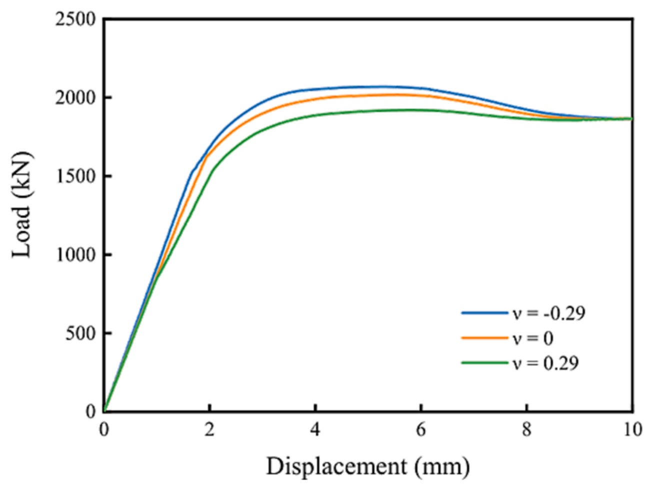

Fig. 13. Load-displacement curves of CFSST composite structures with different Poisson's ratio.

6.2. Influence of thickness of auxetic stainless steel tube

Fig. 14 shows typical load-displacement curves of CFASST FE models with various tube thicknesses, where the simulation results of the models with the constant diameter of the concrete are plotted. The thickness of the stainless steel tube is set to 8 mm, 10 mm and 12 mm, respectively. It is worth noting that the selected thicknesses of the ASST have no appreciable effect on the ASST and CFASST composite structures' auxetic behaviour [58]. It can be seen that increasing the thickness of the stainless tube leads to an increase in load-carrying capacity. This is because of the combined effects that include the strain-hardening behaviour of the stainless steel tube, thickness, and the confinement effect of steel tubes on concrete. In a word, it can be concluded that increasing the thickness of ASST makes contributions to the compressive resistance of CFASST composite structures.

6.3. Confinement effect of stainless steel tube

So as to illustrate the outer stainless steel tube confinement effect, numerical simulation is used to analyze the restraint effect of three groups of stainless steel tube on concrete. The axial strength enhancement factor C is defined as follow:

$$C = \frac{N_{cfs}}{N_c + N_s} \tag{13}$$

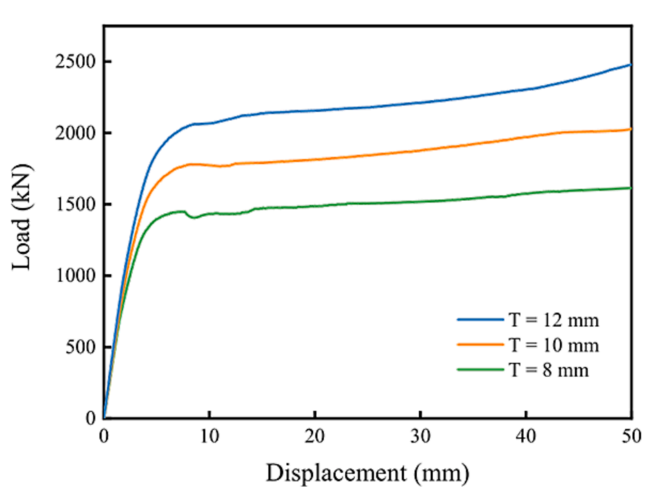

Fig. 14. Load-displacement curves of CFASST composite structures with different thickness.

Table 5
List of CFSST composite structures value.

Specimen label	N_c(kN)	N_s(kN)	N_{cfs}(kN)	C
CFTSST	398.8	1051.1	1910.1	1.32
CFNSST	398.8	627.9	1663.4	1.62
CFASST	398.8	422.1	1779.5	2.32

where the N_{cfs} is the peak load of the FE models for CFSST composite structures, N_c is the ultimate load of the FE models for concrete, N_s is the peak load of the FE models for stainless steel tube. The values of the calculated C are shown in Table 5. It can be seen that all results are greater than 1 and CFASST composite structures have the highest C value. It indicated that outer ASSTs have more confinement effect on the concrete and is extremely important to improve the structural strength. Besides, the load bearing capacity of auxetic tube is relatively low and is difficult for acting as carrying components in the form of pure tubes. Therefore, filling concrete into the auxetic tube is an effective method in engineering applications by making full use of the confinement effect of ASST.

7. Conclusions

In this work, a novel concrete-filled auxetic stainless steel tube (CFASST) composite structure was proposed. Experimental and numerical studies of the CFASST composite structures on compressive performance have been carried out. Three groups of concrete-filled stainless steel tube (CFSST) specimens, including CFASST composite structures, concrete-filled non-auxetic stainless steel tube (CFNSST) composite structures and concrete-filled traditional stainless steel tube (CFTSST) composite structures, were tested and finite element models were established. Finally, parametric analyses were conducted to explore the influence of geometric properties and Poisson's ratio on compressive properties and confinement effect based on validated finite element models. Based on the results, the main conclusions can be summarized as follows:

(1) The axial strength enhancement factor C indicates that CFASST composite structures possess an excellent confinement effect due to the active restraint effect of the auxetic tube on concrete.
(2) Compared with the CFTSST composite structures, CFASST and CFNSST composite structures do not exhibit outward local buckling in their failure modes but show a continuous expansion deformation. The later deformation of CFASST composite structures is less than that of CFNSST composite structures. Overall, CFASST composite structures possess high deformability after entering the plastic stage.
(3) The CFTSST composite structures possess higher compressive stiffness and peak load. However, CFASST composite structures' peak strain is greater than that of other CFSST composite structures and their load-bearing capacity does not show a declining trend which means CFASST composite structures possess better ductility and toughness.
(4) When the Poisson's ratio of the tube decreases, CFSST composite structures' stiffness and peak load will increase accordingly. It shows that decreasing Poisson's ratio is an effective method to improve the compressive performance of CFSST composite structures.
(5) The thickness of the auxetic stainless steel tube (ASST) has an important influence on compressive bearing capacity. As the thickness of the SST increases, the load-bearing capacity and confinement effect will be significantly improved.

Overall, a novel CFASST composite structure was designed, fabricated and investigated for the first time. The compressive performance of the CFASST composite structures was examined through experiment and numerical methods. CFASST columns realized the active restraint effect of the auxetic tube on the concrete from the initial elastic stage, thereby improving the interface bonding effect. The proposed CFASST column possesses certain performance and deformation advantages, such as excellent confinement effect, better ductility and toughness, avoid outward local buckling, however, the negative aspects still need to be further investigated. It is worthwhile to take advantage of topology optimization to design auxetic tubes and advanced manufacturing technology to fabricate CFASST composite structures with preferred performance. The novel CFASST composite structure has the potential to be used as superior buffer composites for uniaxial crushing, resulting in possible application prospects in the fields of structure protection and construction engineering due to its long load platform phase.

CRediT authorship contribution statement

Chen Luo: Conceptualization, Methodology, Software, Writing – review & editing, Investigation. **Xin Ren:** Project administration, Conceptualization, Methodology, Supervision, Funding acquisition, Resources , Writing - review & editing. **Dong Han:** Investigation, Validation. **Xue Gang Zhang:** Investigation, Validation. **Ru Zhong:** Investigation, Validation. **Xiang Yu Zhang:** Validation. **Yi Min Xie:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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